

Kanika Prajapati

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KEY SKILLS

API Development | System Architecture | Database Management | Performance Optimization | Testing & Automation | Data Structures & Algorithms | Object-Oriented Programming | Distributed Systems | SDLC | Unit Testing | Cloud Computing | Troubleshooting & Problem-Solving | Cross-Team Collaboration | Agile Development

EXPERIENCE

• Software Engineer 1, Intuit

Aug, 2025 - July, 2026

- Increased **load handling capacity** by **27%** with the **AWS S3** caching layer, eliminating **DynamoDB hot-partition** bottlenecks.
- Migrated **5,000+** legacy web pages to GWP with **100% feature parity**, driving **SEO adoption** from **0% to 85%** and **boosting page load performance** by **95%**.
- Designed **LLDs** and **HLDs** for 3 key migration features, shortening design-review cycles and enabling high-availability rollout.
- Implemented **SQS/Scheduler-triggered Lambda** workflows to generate **S3 snapshots**, **reducing page failures** by **93%**.
- Automated content-variant generation via a new **IXP/PZN API**, **eliminating 95% of manual preview steps**.
- Automated QA by building an **AI-powered visual-regression tool**, **reducing manual effort** by **95%**.
- Conducted **Gatling load and chaos testing** across multiple microservices, uncovering critical reliability issues pre-release.
- Sustained uptime across 3 services by **monitoring and deploying Kubernetes pods via Argo CD** during **on-call rotations**.

• Software Engineering Intern, Intuit

May, 2024 - Aug, 2024

- **Automated** an onboarding workflow using **Spring Boot**, **reducing SLA** from **2-3 days to near real-time**.
- Built **status-tracking APIs** with **event-driven state transitions**, improving request visibility.
- Optimized **DynamoDB query** through **global secondary index** design for retrieving pending requests.
- Developed **JUnit 5** and **Mockito** unit tests, achieving **100% code coverage** across critical service modules.

EDUCATION

B.Tech. Computer Science and Engineering, Indian Institute of Technology (BHU), Varanasi **CGPA: 8.83**, 2021-2025

TECHNICAL SKILLS

Languages:	C++, Python, SQL, Java
Backend:	Spring Boot, REST APIs, GraphQL, Microservices, CI/CD
Cloud and Databases:	AWS (Lambda, S3, DynamoDB, SQS, SNS, EventBridge Scheduler)
Tools:	Git, Linux, Maven, Jenkins

PROJECTS

• Coding Interview Agent

- Built a **multi-agent LangGraph** workflow with specialized agents for hints, evaluation, and complexity analysis.
- Implemented **ChromaDB-based RAG** over DSA concepts and coding patterns for **context-aware feedback**.
- Integrated **Ollama API** for local LLM inference and agent orchestration.

Exposure: LangGraph, Ollama API, ChromaDB, Multi-Agent Workflows

• Task Scheduling using Archimedes Optimization Algorithm

- Executed AOA in **CloudSim** for dynamic task allocation in complex **distributed systems**.
- **Reduced scheduling makespan by 30%** by optimizing virtual machine (VM) utilization.
- **Co-authored a research paper** evaluating AOA-based workflow scheduling.

Exposure: CloudSim, Cloud Computing, Distributed Systems, Resource Scheduling

• Predicting metallic glass forming ability through characteristic temperatures

- Built **stacking-based meta-models** and **ensemble classifiers** for metallic glass forming ability prediction.
- Applied **ensemble methods** over the classification dataset and obtained an **F1 score of 0.84**.

Exposure: Python, Scikit-learn, Ensemble Learning (Random Forest), Stacking Meta-models

• Inventory Management System

- Built a **Spring Boot** backend for inventory and order management across multiple warehouses.
- Designed **REST APIs** for stock updates and order processing with **MySQL** integration.
- Implemented low-stock alerts and optimized database queries, **reducing response time by 35%**.

Exposure: Java, Spring Boot, MySQL, RESTful APIs

AWARDS AND ACHIEVEMENTS

- Secured 98.9 percentile and All India Rank **2065 (OBC-NCL)** in **JEE (MAINS) 2021**.
- Secured All India Rank **4281** in **JEE (Advanced) 2021** among more than 2,00,000 shortlisted students.
- Achieved **Specialist** rating on Codeforces and solved **500+ coding questions** on all platforms.